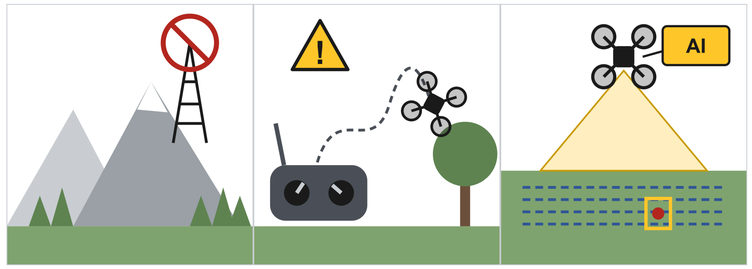
XX-000

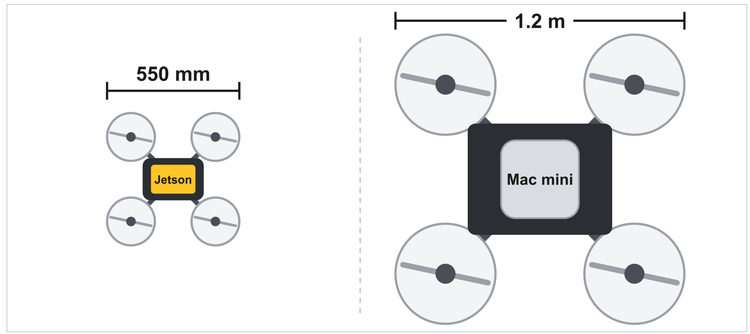
KESTREL: Just Say Who's Missing

Shiva Shrestha and Honghui Xu

The Problem



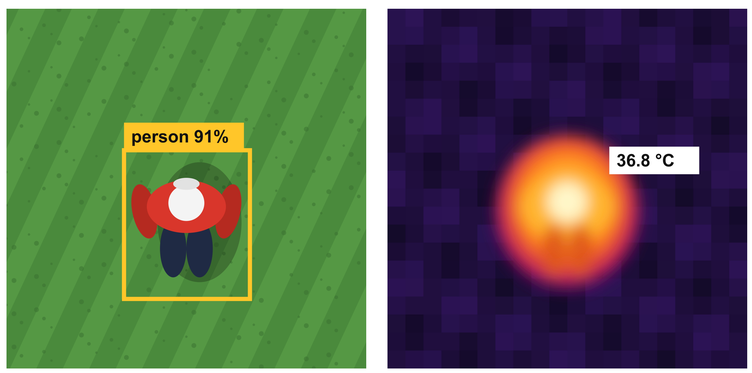
Two Drone Models



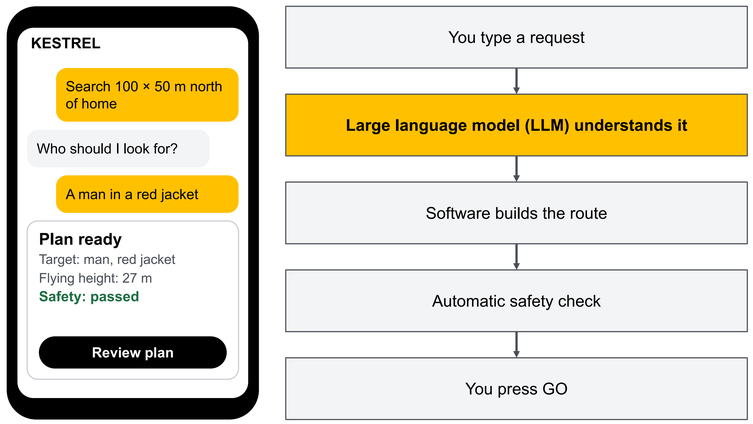
Compact, 550 mm
Jetson · Gemma 4 E2B

Large, 1.2 m
Mac mini · Qwen3.6-35B-A3B

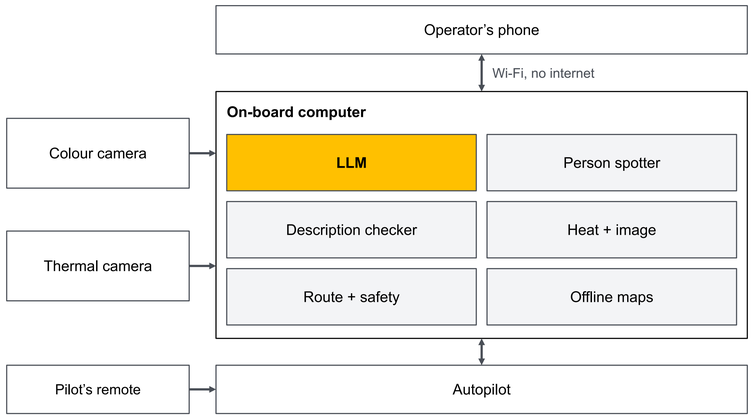
Finding the Person



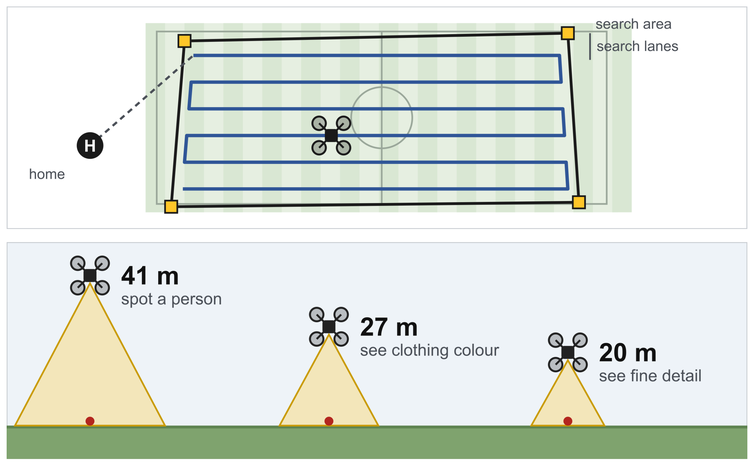
Plan a Search by Chatting



System Overview



Search Pattern and Height

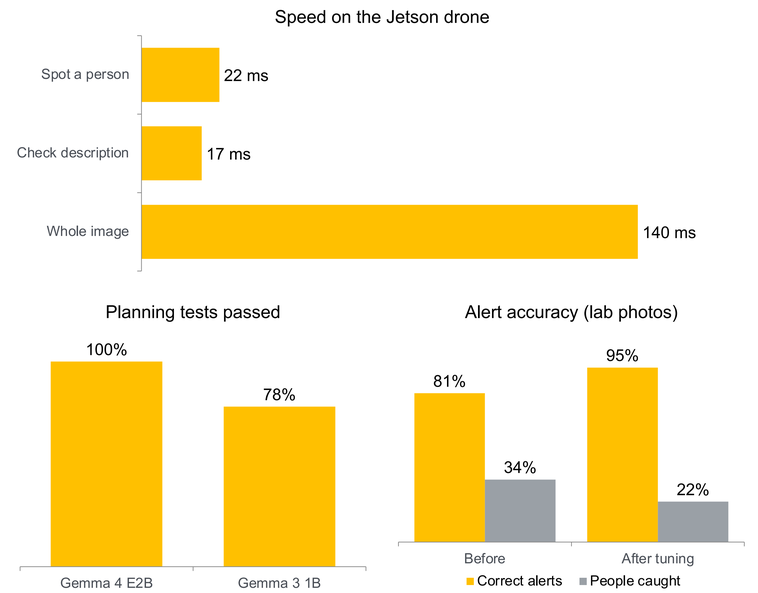


Automatic Safety Checks

- ✓ GPS locked
- ✓ Virtual fence
- ✓ Airspace approval
- ✓ Autopilot self-test
- ✓ Cameras and AI
- ✓ Battery level
- ✓ 400 ft limit
- ✓ Pilot's remote
- ✓ Route verified
- ✓ Legal checklist

FAA Part 107 · licensed pilot can take over

Results



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